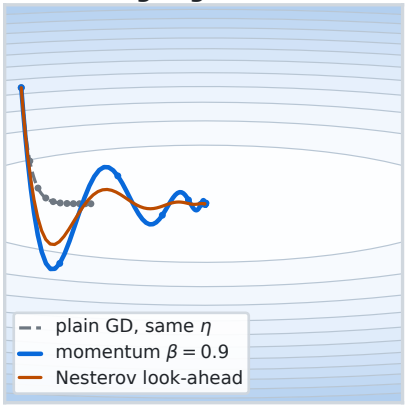
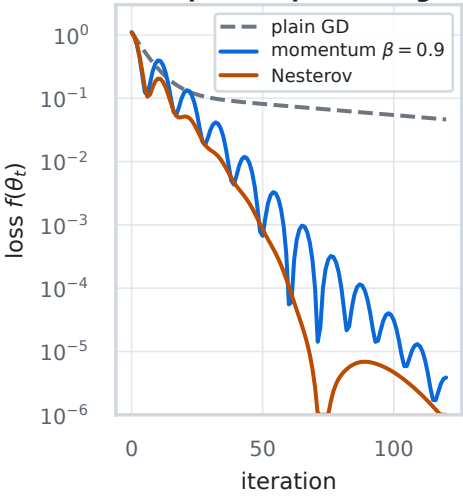
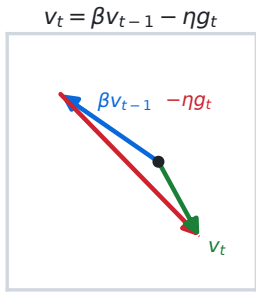

(a) the zig-zag is filtered out
plain GD, same η
momentum $\beta = 0.9$
Nesterov look-ahead


(b) same step size: speed along the floor
plain GD
momentum $\beta = 0.9$
Nesterov
loss $f(\theta_t)$
10^{0}
10^{-1}
10^{-2}
10^{-3}
10^{-4}
10^{-5}
10^{-6}
0
50
100
iteration


$v_t = \beta v_{t-1} - \eta g_t$
βv_{t-1}
$-\eta g_t$
v_t